

FABRID-IDS: Federated Alert-Budget Reallocation for Heterogeneous IoT Intrusion Detection

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Abstract

A frozen federated anomaly scorer can rank attacks well while still producing unreliable operating thresholds across heterogeneous clients. FABRID-IDS treats post-training threshold selection as federated alert-budget reallocation: client-specific nominal false-positive-rate (FPR) targets are assigned under a shared weighted budget and then calibrated independently on local benign data. The primary evaluation uses nine N-BaIoT devices, ten detector seeds, and five nominal budgets; CIC IoT-DIAD provides a secondary protocol stress test with different clients, features, and a different scorer. At $B = 0.001$, unrestricted Macro and Tail allocations realize a federation FPR of **0.00170**, or **1.696B**, despite satisfying the nominal allocation constraint. Conservative calibration reduces the realized FPR to **0.00036** but lowers macro recall from **0.3778** to **0.2445** and does not restore worst-client recall. Mean AUROC and AUPRC are **0.9995** and **0.9999**, yet FABRID-family minimum-client recall remains zero across the evaluated N-BaIoT budgets. Thus, satisfying the nominal federation-level alert-budget constraint does not guarantee the realized FPR or client-level detection behavior after finite local calibration, even when score ranking is near perfect.

Keywords: federated intrusion detection, Internet of Things, alert-budget reallocation, false-positive-rate constraint, threshold calibration

1 Introduction

Federated IoT detection distributes model training across clients, but deployment still requires a threshold that converts anomaly scores into alerts [1–3]. Shared and client-specific thresholding have been studied [4–6], yet neither determines how a federation-wide FPR budget should be divided across heterogeneous clients. With a frozen scorer, differences in benign-score distributions and attack-recall curves can produce uneven false-alarm and detection-utility outcomes across clients.

FABRID-IDS treats this as alert-budget reallocation. Given a fixed scorer, it selects client-specific nominal rates α_k under the federation constraint $\sum_k w_k \alpha_k \leq B$, recalibrates thresholds on disjoint benign data, and evaluates them on held-out records. B is a nominal federation-level FPR budget, not a realized-FPR guarantee.

The contributions are federation-constrained alert-budget allocation, a frozen-scorer protocol separating allocation, calibration, and testing, and paired N-BaIoT evidence that nominal compliance need not preserve realized FPR or worst-client recall.

The rest of the paper covers related work, methods, results, and implications.

2 Related work and scope

Federated IoT IDS research has mainly focused on detector training [1–3, 7], while threshold regulation and distributed anomaly detection address related operating-point questions [8, 9]. Laridi et al. [4] aggregate client summary statistics into one global anomaly threshold. Asiri et al. [5] compute local 95th-percentile benign reconstruction-error thresholds and average them into one global threshold. Fed-DTCN [6] uses held-out benign calibration data to select client-specific thresholds for target local FPRs. These designs neither jointly allocate client-specific nominal FPR targets under one federation-level budget nor formulate a finite-sample guarantee for the resulting realized federation FPR.

Rate-constrained and Neyman-Pearson methods address explicit error-rate control [10, 11], while federated and conformal calibration provide finite-sample control under stated sampling or exchangeability assumptions [12, 13]. FABRID-IDS jointly selects client-specific nominal rates under $\sum_k w_k \alpha_k \leq B$, recalibrates thresholds on disjoint benign data, and evaluates realized federation and client-level FPR. The federation-level allocation formulation is the contribution; CVaR, order statistics, mixed-integer optimization, and finite-sample calibration are established components. Because N-BaIoT is split by released source order, the splits establish neither deployment chronology nor exchangeability.

3 FABRID-IDS

3.1 Protocol and utility

For K clients, a frozen scorer $s_k(\cdot)$ produces anomaly scores. Released source order defines benign training (50%), frontier (20%), final-calibration (10%), and test (20%) splits; the initial 20% of each retained attack subtype forms the validation split and the

remainder forms the test split. The pipeline proceeds through training, frozen scoring, frontier utility estimation, allocation, final calibration, and testing; final-calibration and test benign records are not allocation inputs.

For ascending benign scores $x_{(1)}, \dots, x_{(n)}$ and nominal target α , threshold calibration uses a standard order-statistic construction [13]

$$\tau(\alpha) = x_{(r)}, \quad r = \lceil (n+1)(1-\alpha) \rceil, \quad (1)$$

with $\tau = +\infty$ if $\alpha \leq 0$, $n = 0$, or $r > n$; alerts require $s > \tau$. Frontier utility $u_k(\alpha)$ is the mean recall across attack subtypes. Allocation utility is estimated from labeled attack validation data; only the subsequent final threshold calibration uses benign data. The evaluated procedure is therefore attack-informed alert-budget reallocation rather than label-free deployment adaptation. Eligible subtypes require at least 50 validation rows, clients require at least 200 rows across two subtypes, and the 207-point grid spans $[0, 0.05]$ and includes every evaluated budget.

3.2 Allocation and calibration

FABRID-IDS reallocates the nominal alert budget by selecting client-specific grid rates subject to

$$\sum_{k=1}^K w_k \alpha_k \leq B. \quad (2)$$

Primary N-BaIoT results use $w_k = 1/K$, so nominal and realized federation FPR are both equal-client averages. One-hot mixed-integer programs (HiGHS) maximize mean u_k (Macro) or lower-tail CVaR at quantile 0.25 (Tail) [14]. For nine clients, the Tail objective averages the three lowest utilities. Tail optimizes lower-tail CVaR at the stated quantile, whereas minimum-client recall is reported separately as a stricter worst-client stress endpoint. Candidate frontier rates below twice the order-statistic transition are excluded.

Final thresholds are recalibrated with Eq. (1). The conservative intervention selects the smallest rank satisfying the binomial calibration construction [13]

$$r_{.95} = \min\{r : \Pr[\text{Binomial}(n, 1-\alpha) \leq r-1] \geq 0.95\}. \quad (3)$$

The resulting threshold retains the strict $s > \tau$ alert rule. This rank adjustment is used as a finite-sample sensitivity intervention rather than as a deployment guarantee under the source-order split. Its nominal 95% interpretation requires the calibration data and future benign observations to satisfy the stated i.i.d. assumption.

With strict alerts, $\alpha < 2/(n+1)$ gives either $r > n$ or a maximum-score threshold and zero calibration alerts; the smallest nonzero empirical FPR is $1/n$ (absent ties). Rates below this transition cannot generate a nonzero empirical calibration FPR under strict thresholding; excluding them prevents the optimizer from assigning nominal budget to grid points that are empirically indistinguishable from a zero-alert calibration point. Ecobee has $n = 1311$ on the final-calibration split, so targets below 0.001524 calibrate to zero alerts. This calculation explains why sufficiently small allocated rates

produce no calibration alerts, but it does not explain the subsequent held-out FPR shift.

3.3 Frozen scorer

The primary scorer is the maximum of frozen frontier-reference percentile ranks for autoencoder reconstruction error and the mean of the jitter-based temporal feature `HH_jit_L0.1_mean`, denoted f_{75} . The f_{75} candidate and scorer ablations were predeclared for development-only selection. The autoencoder uses [64,16] layers, ten rounds of three local epochs, row-count-weighted averaging, Adam (0.001), batch size 64, MSE, and per-client training-split normalization; ten detector seeds are the paired replication unit.

4 Experimental methodology

The primary experiment uses nine N-BaIoT devices and five prespecified budgets from $B = 0.001$ through $B = 0.020$ (Table 1). Macro recall is the mean of client-level macro recalls, each computed over eligible attack subtypes; minimum-client recall is their minimum.

Prespecified contrasts compare Macro with Equal-FPR on macro recall and Tail with Equal-FPR on minimum-client recall using ten paired seeds, exact sign-flip tests, 50,000-resample paired-bootstrap CIs, and Holm correction over each five-budget family. CIC IoT-DIAD provides a 19-client secondary protocol stress test using 57 features and reconstruction error alone. Because the client population, feature representation, and scorer all change, the experiment is not used for cross-dataset mechanism attribution.

5 Results

5.1 Nominal and realized operating points

At $B = 0.001$, unrestricted Macro and Tail each realize federation FPR 0.00170 ($1.696B$), whereas their conservative variants realize 0.00036 ($0.358B$) and macro recall falls from 0.3778 to 0.2445. At $B = 0.005$, Macro realizes $0.706B$ with recall 0.4644, while Tail realizes $1.290B$ with recall 0.4977. These same- B comparisons hold nominal alert-budget expenditure fixed rather than realized FPR. The optimizer constrains $\sum_k w_k \alpha_k$, whereas evaluation measures the FPR produced after independent local calibration. The observed divergence shows that nominal budget compliance can coexist with federation-level overshoot or undershoot and different detection utility.

5.2 Tail objective and worst-client behavior

Tail optimizes lower-tail CVaR, but minimum-client recall is zero for every FAB-RID policy-budget cell and cannot distinguish unrestricted Macro and Tail. `Dan-mini_Doorbell` is the persistent worst-client case: its macro attack recall, final-calibration benign FPR, and held-out benign FPR are all zero in all 100 unrestricted-policy-seed-budget cells. With $n = 4955$ benign final-calibration records, its zero-alert

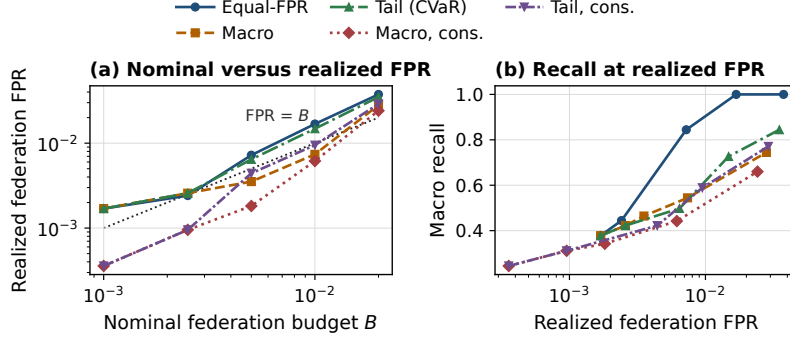


Fig. 1 N-BaIoT means across ten detector seeds. (a) Nominal federation alert budget B against realized equal-client federation FPR; the diagonal marks equality. (b) Macro recall at each observed realized federation FPR. Lines connect evaluated operating points only.

transition is $2/(n+1) = 0.000404$; its allocated-rate range crosses this finite-sample transition.

Ecobee illustrates a different calibration/test discrepancy. At $B = 0.020$, Macro assigns mean $\alpha_k = 0.001164$, below $2/(1311+1)$; its final-calibration FPR is therefore zero, while mean held-out benign FPR is 0.006672 and recall is 0.6000. Finite calibration resolution explains the zero-alert calibration point, but the experiment cannot separate that effect from distributional differences between the source-order final-calibration and test partitions.

5.3 Primary contrasts and ranking ceiling

Macro and Equal-FPR are identical at $B = 0.001$; their -0.0222 difference at $B = 0.0025$ is not statistically distinguishable from zero, and Macro recall is lower at larger

Table 1 N-BaIoT means across ten detector seeds: macro recall and realized federation FPR divided by nominal budget. Minimum-client recall is zero for every FABRID-family policy–budget cell.

B	Equal-FPR	Macro	Tail	Macro, cons.	Tail, cons.
<i>(a) Macro recall</i>					
0.0010	0.3778	0.3778	0.3778	0.2445	0.2445
0.0025	0.4445	0.4222	0.4222	0.3111	0.3111
0.0050	0.8444	0.4644	0.4977	0.3422	0.4222
0.0100	0.9999	0.5444	0.7266	0.4422	0.5889
0.0200	1.0000	0.7444	0.8444	0.6600	0.7711
<i>(b) Realized FPR/B</i>					
0.0010	1.70	1.70	1.70	0.36	0.36
0.0025	0.97	1.03	1.03	0.38	0.38
0.0050	1.45	0.71	1.29	0.36	0.88
0.0100	1.68	0.74	1.48	0.62	0.95
0.0200	1.88	1.40	1.75	1.21	1.45

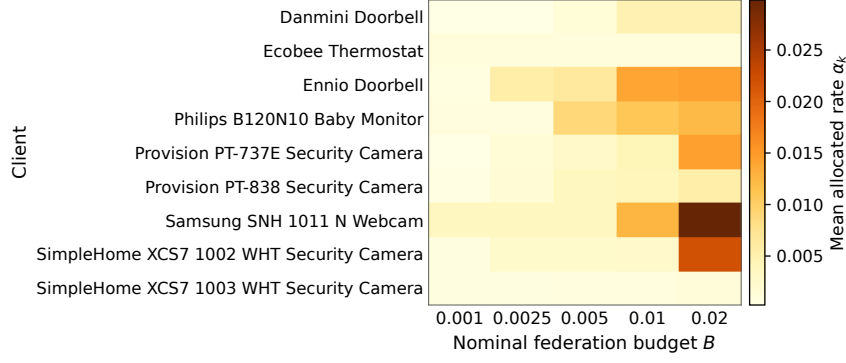


Fig. 2 Mean Macro-policy client-specific nominal rates α_k across ten N-BaIoT detector seeds. Each column satisfies $\sum_k w_k \alpha_k \leq B$. Across 500 resamples of the frontier and attack-validation scores, selected client rates change in 77.8–84.9% of replicates across budgets; the heatmap therefore summarizes mean allocation patterns rather than stable client-specific operating points.

Table 2 Prespecified paired seed-level contrasts ($n = 10$): 50,000 paired-bootstrap CIs and Holm-adjusted exact paired sign-flip p values over the five-budget family.

B	Mean Δ	95% CI	Holm p
<i>Macro – Equal-FPR, macro recall</i>			
0.0010	0.0000	[0.0000, 0.0000]	1.0000
0.0025	−0.0222	[−0.0556, 0.0111]	0.3594
0.0050	−0.3800	[−0.4133, −0.3466]	0.0098
0.0100	−0.4555	[−0.5000, −0.4111]	0.0098
0.0200	−0.2556	[−0.2889, −0.2222]	0.0098
<i>Tail – Equal-FPR, minimum-client recall</i>			
0.0010	0.0000	[0.0000, 0.0000]	1.0000
0.0025	0.0000	[0.0000, 0.0000]	1.0000
0.0050	0.0000	[0.0000, 0.0000]	1.0000
0.0100	−0.9998	[−0.9998, −0.9998]	0.0098
0.0200	−0.9999	[−0.9999, −0.9999]	0.0098

budgets. Tail and Equal-FPR have identical minimum-client recall through $B = 0.005$; Equal-FPR is higher at the two larger budgets (Table 2).

Because allocation changes thresholds rather than anomaly scores, AUROC and AUPRC are policy-invariant by construction. Their mean values are 0.9995 and 0.9999, respectively, yet FABRID-family minimum-client recall remains zero across all five N-BaIoT budgets. Near-perfect ranking metrics therefore do not imply reliable worst-client behavior at the evaluated low-FPR operating points.

5.4 Secondary protocol stress test

CIC IoT-DIAD provides a secondary protocol stress test. Across the five nominal budgets, macro recall ranges from 0.3614 to 0.5016, realized federation FPR from

Table 3 CIC IoT-DIAD secondary protocol stress test: 19 clients, ten seeds, equal-client weighting, and reconstruction-error scoring.

B	Macro recall	Minimum-client recall	Realized FPR
0.0010	0.3614	0.0000	0.0098
0.0025	0.3970	0.0000	0.0118
0.0050	0.4318	0.0007	0.0160
0.0100	0.4602	0.0089	0.0213
0.0200	0.5016	0.0251	0.0311

0.0098 to 0.0311, and minimum-client recall from 0.0000 to 0.0251. The nominal-to-realized mismatch therefore remains visible under this second protocol, but different clients, features, and reconstruction-error-only scoring prevent attribution to the same mechanism as in N-BaIoT.

6 Discussion

On N-BaIoT, nominal budget feasibility does not imply realized budget fidelity. At $B = 0.001$, unrestricted policies realize $1.696B$; conservative calibration lowers realized FPR and recall without recovering Danmini Doorbell detection. Equal-FPR achieves higher macro recall at several budgets, so the evidence does not support uniform superiority of any allocation policy.

Finite calibration granularity can force zero-alert points, as for Ecobee, but the source-order split cannot separate this effect from calibration/test distribution differences and does not establish temporal drift. Attack-informed utility shows limited transfer in family-disjoint stress tests, which remain within-benchmark rather than evidence of OOD generalization.

Evidence is limited to one scorer, nine devices, ten seeds, five budgets, and the CIC IoT-DIAD stress test. Privacy, adversarial clients, deployment overhead, and scorer-level remedies are outside scope.

7 Conclusion

FABRID-IDS allocates client-specific nominal FPR targets under a shared federation budget and recalibrates thresholds locally. Nominal compliance can still produce realized FPR mismatch or zero worst-client recall. Limitations include source-order calibration, attack-informed utility, and the nine-device benchmark. Future work should study deployment-ordered calibration, distribution shift, and realized-FPR control under explicit sampling assumptions. Code and artifacts are available at <https://doi.org/10.5281/zenodo.22113172>. The authors declare no competing interests.

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